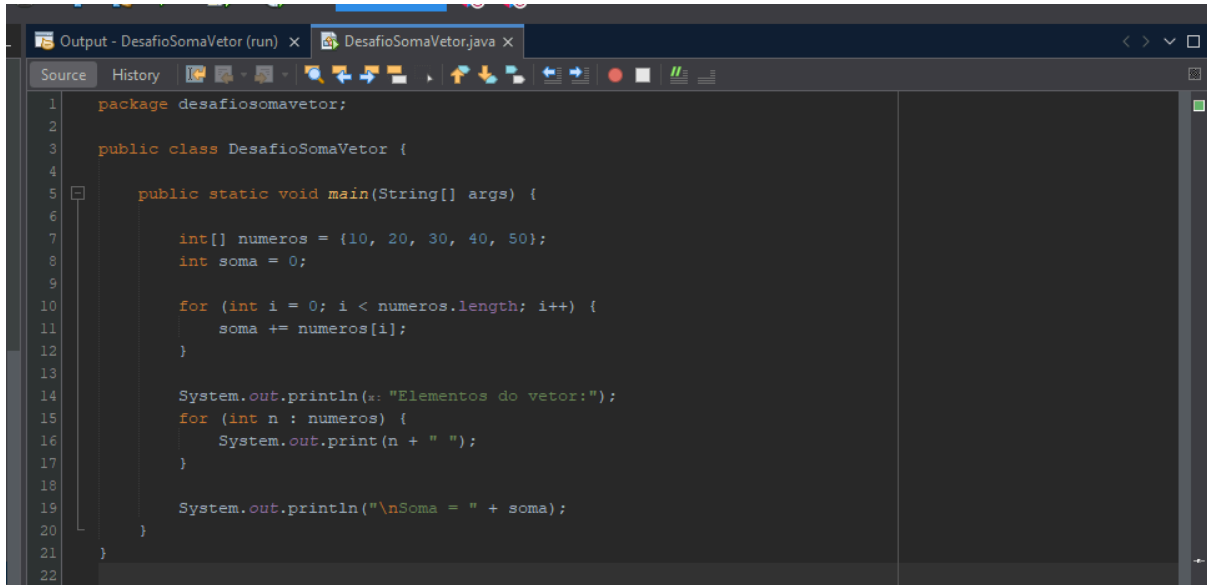
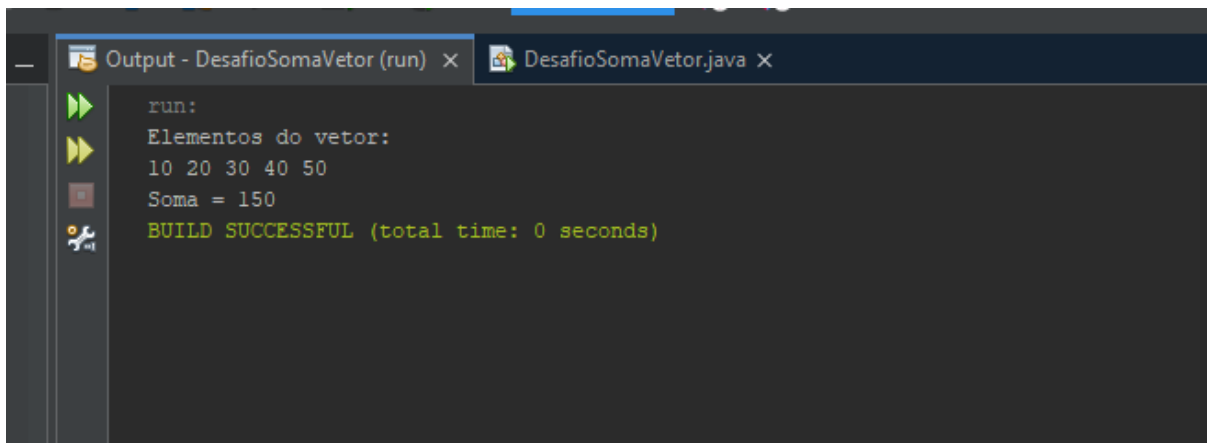


Desafios sobre Vetores e Matrizes em Java

Desafio 1 - Soma de Vetor



```
1 package desafiosomavetor;
2
3 public class DesafioSomaVetor {
4
5     public static void main(String[] args) {
6
7         int[] numeros = {10, 20, 30, 40, 50};
7         int soma = 0;
8
9         for (int i = 0; i < numeros.length; i++) {
10             soma += numeros[i];
11         }
12
13         System.out.println(s: "Elementos do vetor:");
14         for (int n : numeros) {
15             System.out.print(n + " ");
16         }
17
18         System.out.println("\nSoma = " + soma);
19     }
20 }
21
22
```



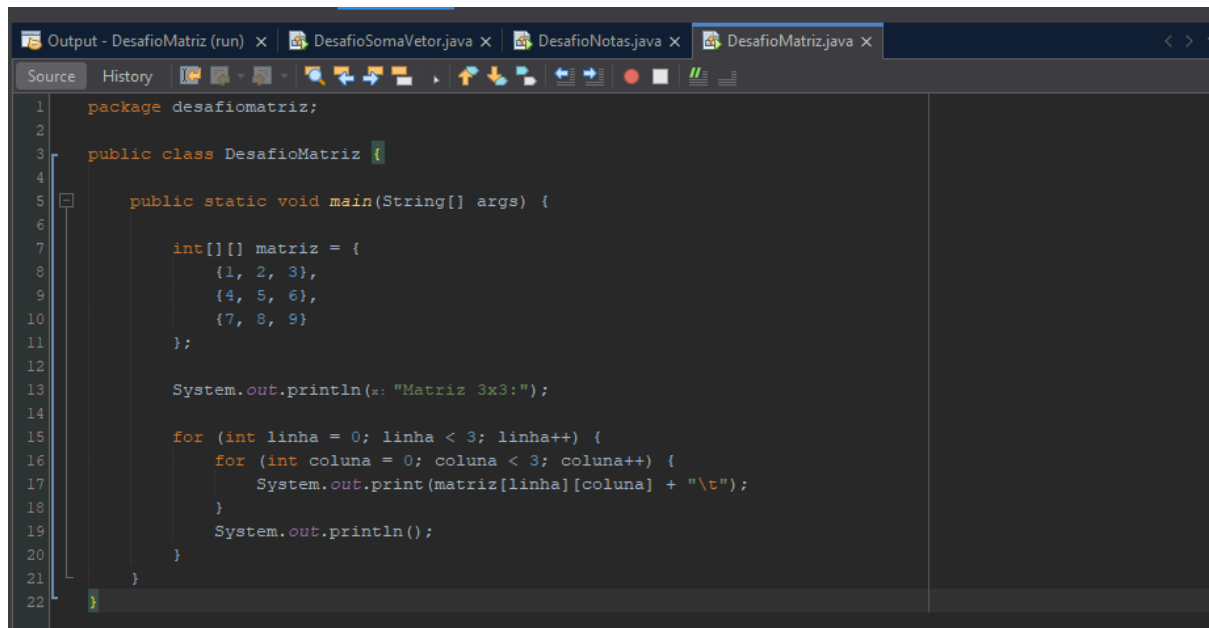
```
run:
Elementos do vetor:
10 20 30 40 50
Soma = 150
BUILD SUCCESSFUL (total time: 0 seconds)
```

Desafio 2 - Notas dos Alunos

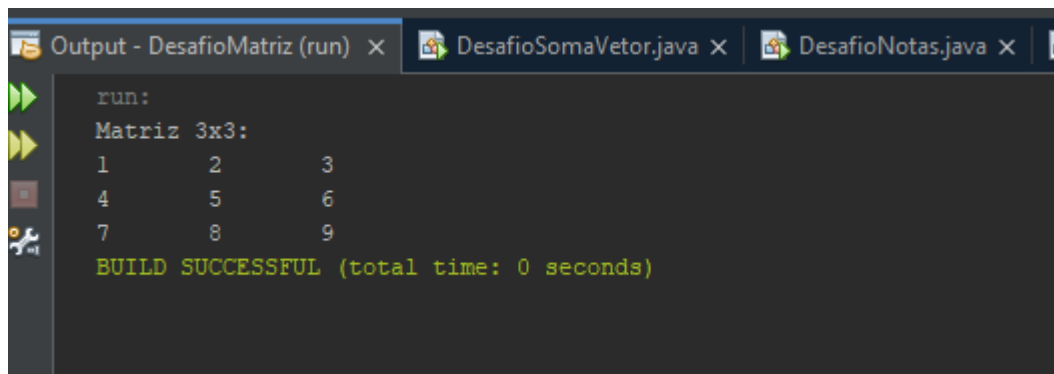
```
Source History
1 package desafio2notas;
2
3 public class DesafioNotas {
4
5     public static void main(String[] args) {
6
7         double[] notas = {8.5, 7.0, 9.0, 6.5, 10.0};
7         double soma = 0;
8
9         for (double nota : notas) {
10             soma += nota;
11         }
12
13         double media = soma / notas.length;
14
15         System.out.println(x: "Notas dos alunos:");
16         for (double nota : notas) {
17             System.out.println(x: nota);
18         }
19
20         System.out.printf(format: "Média = %.2f\n", args: media);
21     }
22 }
23 }
```

```
Output - DesafioNotas (run) x  DesafioSomaVetor.java x  DesafioNotas.java x
run:
Notas dos alunos:
8.5
7.0
9.0
6.5
10.0
Média = 8,20
BUILD SUCCESSFUL (total time: 0 seconds)
```

Desafio 3 - Matriz 3x3

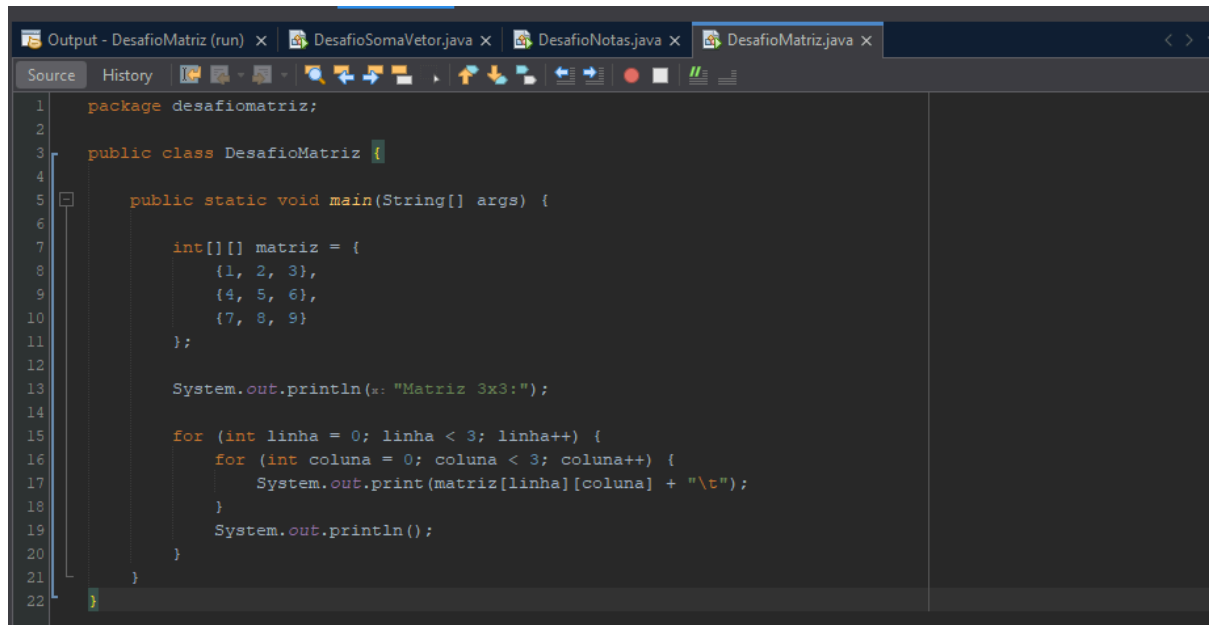


```
1 package desafioMatriz;
2
3 public class DesafioMatriz {
4
5     public static void main(String[] args) {
6
7         int[][] matriz = {
8             {1, 2, 3},
9             {4, 5, 6},
10            {7, 8, 9}
11        };
12
13        System.out.println("Matriz 3x3:");
14
15        for (int linha = 0; linha < 3; linha++) {
16            for (int coluna = 0; coluna < 3; coluna++) {
17                System.out.print(matriz[linha][coluna] + "\t");
18            }
19            System.out.println();
20        }
21    }
22 }
```

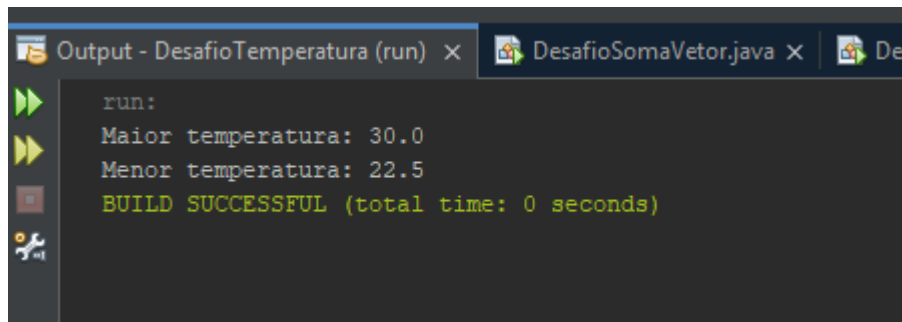


```
run:
Matriz 3x3:
1      2      3
4      5      6
7      8      9
BUILD SUCCESSFUL (total time: 0 seconds)
```

Desafio 4 - Temperaturas



```
1 package desafiomatriz;
2
3 public class DesafioMatriz {
4
5     public static void main(String[] args) {
6
7         int[][] matriz = {
8             {1, 2, 3},
9             {4, 5, 6},
10            {7, 8, 9}
11        };
12
13        System.out.println("Matriz 3x3:");
14
15        for (int linha = 0; linha < 3; linha++) {
16            for (int coluna = 0; coluna < 3; coluna++) {
17                System.out.print(matriz[linha][coluna] + "\t");
18            }
19            System.out.println();
20        }
21    }
22 }
```



```
run:
Maior temperatura: 30.0
Menor temperatura: 22.5
BUILD SUCCESSFUL (total time: 0 seconds)
```

Desafio 5 - Tabuada com Matriz

```
1 package desafioTabuada;
2
3 public class DesafioTabuada {
4
5     public static void main(String[] args) {
6
7         int[][] tabuada = new int[10][10];
8
9         for (int i = 0; i < 10; i++) {
10             for (int j = 0; j < 10; j++) {
11                 tabuada[i][j] = (i + 1) * (j + 1);
12             }
13         }
14
15         System.out.println("Tabuada (1 a 10):");
16
17         for (int i = 0; i < 10; i++) {
18             for (int j = 0; j < 10; j++) {
19                 System.out.printf(format: "%4d", tabuada[i][j]);
20             }
21             System.out.println();
22         }
23     }
24 }
```

```
run:
Tabuada (1 a 10):
1  2  3  4  5  6  7  8  9 10
2  4  6  8 10 12 14 16 18 20
3  6  9 12 15 18 21 24 27 30
4  8 12 16 20 24 28 32 36 40
5 10 15 20 25 30 35 40 45 50
6 12 18 24 30 36 42 48 54 60
7 14 21 28 35 42 49 56 63 70
8 16 24 32 40 48 56 64 72 80
9 18 27 36 45 54 63 72 81 90
10 20 30 40 50 60 70 80 90 100
BUILD SUCCESSFUL (total time: 0 seconds)
```